

Paper Summary

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Paper : How to Read a Paper — S. Keshav

This paper felt like genuinely useful advice for someone who is just beginning to explore research. Instead of treating reading a research paper as something that must always be done from the first page to the last, Keshav proposes a three-pass approach. The first pass is a quick scan to understand the paper's category, context, correctness, contributions, and clarity. The second pass focuses on understanding the main ideas and evidence while avoiding unnecessary details. The third pass is reserved for papers that require deep understanding, where the reader attempts to virtually re-implement the work and challenge its assumptions.

What I found most useful was the idea that the effort spent reading a paper should depend on how important that paper is to me. I previously thought that understanding a paper meant eventually understanding every detail, but this approach makes the process much more practical. If a paper is only remotely relevant, a first pass may be enough; if it is important to my work, I can invest progressively more time. This seems especially useful because I expect to encounter a large number of papers while working on both my TY project and my chess engine research.

I also found the literature survey process particularly valuable. Starting with a few recent papers, following their related work and shared citations, and identifying recurring researchers and key papers provides a systematic way to explore an unfamiliar research area rather than searching randomly.

The next time I read a research paper, I plan to begin with the first pass and ask myself what I actually need from the paper before deciding how deeply to read it. For papers directly relevant to my projects, I will move to the second or third pass and actively question the assumptions, methodology, and results. I think this will make my reading more efficient while also helping me become more critical and curious as a researcher.